Interactive Multilevel Fusion With Dynamic Bounding Boxes for Small Object Detection in Clutter

Chunzhi Li, Shengyang Su, Xiaohua Chen, Yuan Zhang, and Jiaxin Li, *Member, IEEE*

***Abstract*—Detecting small targets in remote sensing imagery remains a critical challenge due to the inherent limitations of low-resolution targets and the frequent clustering of small targets. To address these challenges, we propose an integrated set of improvements to the You Only Look Once (*YOLO*)v8 framework, leveraging cross-connection mechanisms to construct a robust small target detector. Specifically, we first introduce a Multiscale Feature Enhancement Module (*MFEM*) that utilizes multiscale depth-separable convolutions combined with a cross-connection strategy. This design effectively enriches feature representations, significantly improving feature extraction for small targets while maintaining detection performance for medium to large targets. Second, we develop a multilayer feature fusion neck (*MFFN*) architecture that strategically integrates low-level spatial features into high-level semantic layers. This approach significantly improves detection performance in clustered small target scenarios while optimizing the overall feature fusion process. Third, to overcome the problem of limited adaptability of specialized small target losses in complex environments, we propose a dynamic normalized Wasserstein distance (*DNWD*) loss by combining the advantages of dynamically adjusted normalized Wasserstein distance (*NWD*) and complete intersection-over-union (*CIoU*). Extensive experiments demonstrate that our integrated approach consistently outperforms YOLOv8 baselines across multiple model scales. The proposed framework achieves mAP50 gains of 1.6% (YOLOv8n) and 1.8% (YOLOv8s) on the detection in optical remote sensing image (DIOR) dataset, as well as 2.1% (YOLOv8n) and 2.9% (YOLOv8s) on the VisDrone2019 dataset.**

***Index Terms*—Cross connection, dynamic normalized Wasserstein distance (*DNWD*), multilayer feature, multiscale feature, small object detection (*SOD*).**

I. Introduction

RECENT breakthroughs in aerospace technology have significantly advanced optical remote sensing image analysis [1]. As a major research focus, small object detection (*SOD*) in remote sensing imagery has demonstrated significant value for applications, such as smart city planning, precision agriculture, and environmental monitoring [2], [3], [4]. Because the remote sensing data are typically acquired from high-altitude platforms, the acquired images cover large geographical areas but often represent targets at extremely small scales (<32 $\times$ 32 pixels) with frequent clustering phenomena. The unique characteristics of remote sensing data present unique detection challenges: 1) inherently weak feature representations due to low contrast and resolution; 2) increased detection complexity due to mutual target interference; and 3) increased regression difficulty for accurate bounding box localization [5], [6]. These challenges result in inadequate feature extraction for tiny targets and accurate coordinate regression constrained by minimal target dimensions.

Small targets in remote sensing images typically occupy a tiny fraction of pixels, often appearing as sparse clusters. Due to their limited spatial footprint, they are susceptible to occlusion by background elements or other objects, resulting in inherently weak semantic information [7]. In recent years, there have been significant advances in SOD for remote sensing applications, with particular emphasis on multiscale analysis and attention-based architectures. A notable contribution is PKINet developed by Cai et al. [8], which combines parallel multiscale convolutional operations with a context-aware attention mechanism to enhance both local contextual understanding and multiscale feature representation. Building on this paradigm, Zhang et al. [9] introduced a MENet model by incorporating a global context aggregation module that dynamically weights feature importance, significantly improving the model's ability to discriminate small targets from complex backgrounds. Further developments in this area include Liang's multiscale edge embedding attention (*MEEA*) and Lu's multilevel semantic guidance (*MSG*) [10] modules, which each address two critical challenges: MEEA improves scale adaptability by explicitly incorporating edge features, while MSG optimizes feature hierarchy integration by leveraging high-level semantic guidance.

For bounding box regression, recent innovations have overcome traditional limitations in small object localization. Xu et al. [11] proposed the normalized Wasserstein distance (*NWD*) method, which models bounding boxes as Gaussian distributions and uses Wasserstein distance for similarity

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measurement, effectively addressing the sensitivity issues of intersection-over-union (*IoU*) metrics. At the same time, Zhang and Zhang [12] developed Shape-IoU, which incorporates both spatial overlap and geometric features (aspect ratio, area) to improve localization accuracy for variably shaped small targets.

Recent advances show that robust detection of small targets in remote sensing depends on three key components: 1) sophisticated multiscale feature extraction; 2) adaptive feature fusion strategies; and 3) specialized regression approaches tailored to the unique characteristics of small objects. Among modern detection frameworks, the You Only Look Once (*YOLO*) series has emerged as the dominant choice due to its superior scalability and computational efficiency, especially for remote sensing applications. Several recent YOLO-based methods have further improved SOD. For example, Hui et al. [13] introduced dense residual connection blocks and subpixel convolution while optimizing anchor box sizes via K-means++ clustering. Zhang et al. [14] improved YOLO by integrating a multibranch structure that combines standard and atrous convolutions to improve both local context and receptive field. Zheng et al. [15] proposed a YOLOv8n-based detector with double decoupled downsampling and a coordinate attention-equipped feature pyramid network. In addition, Fan et al. [16] addressed feature interaction degradation through an innovative fusion strategy, resulting in their LUD-YOLO detector.

Despite the development of numerous algorithms to address the challenges of SOD, especially significant advances in multiscale modeling and attention-based architectures, performance remains limited when targets appear in clusters or are obscured by cluttered backgrounds. This limitation stems from three core technical gaps: first, most detection networks fail to preserve fine-grained texture information across scales, resulting in incomplete feature representations; second, conventional feature fusion strategies often overlook the complementary relationships between shallow and deep features, leading to insufficient integration of hierarchical information; and third, the existing bounding box regression losses lack adaptive mechanisms that balance spatial alignment with shape sensitivity. This constrains their robustness to diverse small object geometries. To address these challenges comprehensively, we propose a systematic enhancement framework for YOLOv8.

1) First, to improve the feature representation, we propose a multiscale feature enhancement module (*MFEM*). While inspired by FFCA-YOLO's [14] multibranch atrous convolution structure, our MFEM addresses its limitation in preserving fine-grained details. It combines multiscale depthwise separable convolutions, which capture different texture features through different receptive fields, with a novel cross-information fusion structure. The proposed MFEM simultaneously preserves detailed local information while strengthening interscale correlations, ultimately improving contextual representation.
2) Second, to improve feature fusion for separating the clustered or occluded small targets, we develop a multilayer feature fusion neck (*MFFN*). The architecture strategically incorporates low-level features that preserve critical small target details into the neck fusion pathway, while simultaneously performing adjacent layer fusion to capture rich hierarchical context.
3) Finally, to address the limitations of conventional bounding box regression losses, we analyze the characteristics of standard loss functions and propose an improved solution. NWD improves sensitivity to variations in target shape and scale, while *complete IoU (CIoU)* optimizes center-point alignment and aspect ratio matching between the predicted and ground-truth boxes. However, both NWD and CIoU have inherent drawbacks for SOD tasks. To overcome these limitations, we introduce a hybrid dynamic loss function that intelligently integrates NWD and CIoU. This combination provides a robust and efficient regression loss tailored for SOD. Specifically, we propose the dynamic NWD (*DNWD*), which uses an exponential moving average (*EMA*) mechanism to adaptively adjust the NWD contribution based on real-time regression box statistics. This innovation enables fully automatic and adaptive tuning of the NWD and CIoU weights, eliminating the need for manual tuning while optimizing detection performance.

Our contributions are integrated as follows.

1) We incorporate depthwise separable convolutions at multiple scales, combined with cross-fusion, to enhance feature correlations across scales. This design not only improves the representation of small target features but also captures richer local contextual information, enabling robust detection in cluttered environments.
2) To extract subtle small target cues from complex backgrounds, we introduce the MFFN module, which strategically integrates low-level spatial details with high-level semantic features. Our novel multiscale fusion strategy ensures efficient consolidation and enhancement of discriminative feature information across different network layers.
3) We propose DNWD, an adaptive loss compensation mechanism that uses an EMA to dynamically balance the weights between NWD (for shape/scale sensitivity) and CIoU (for spatial orientation). The proposed DNWD implements a self-adaptive bounding box loss calculation, significantly improving localization accuracy for a variety of small objects.

The remainder of this article is organized as follows. In Section II, we provide a detailed description of our proposed innovations. In Sections III and IV, we show the experimental settings and present compelling experimental results along with comprehensive analyses. Finally, Section V concludes this article.

II. Proposed Method

This work re-engineers the YOLOv8 framework to address two persistent bottlenecks in remote sensing SOD: 1) reliable detection of small objects in low-resolution imagery and 2) accurate bounding box regression for densely distributed targets. As shown in Fig. 1, while maintaining a conventional single-stage detector architecture, our approach introduces three synergetic innovations.

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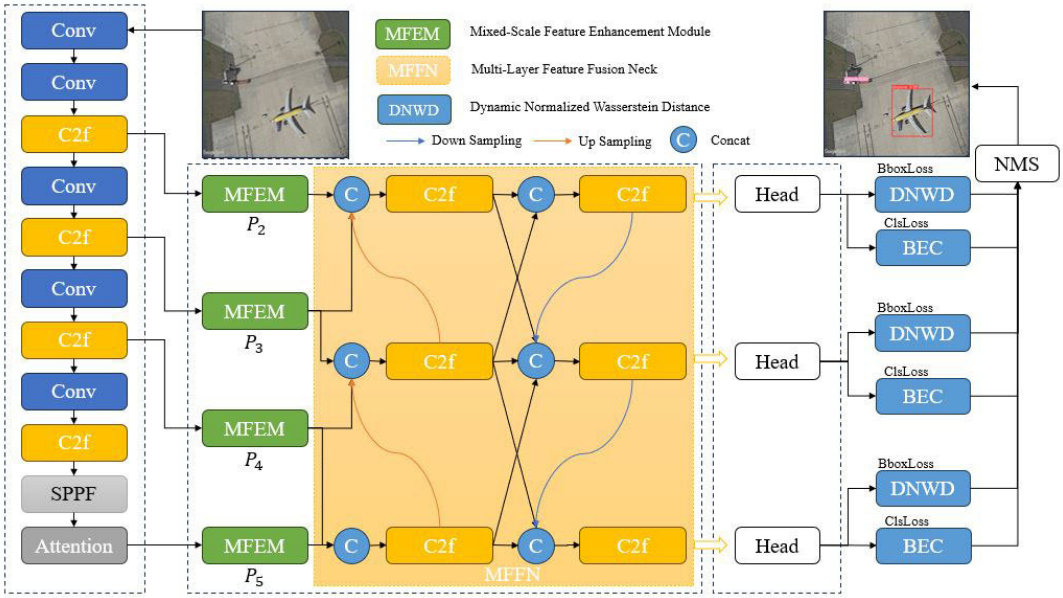


Fig. 1. Comprehensive architecture of our proposed remote sensing SOD model. Here, *conv* represents the convolution operation, and C2f is the feature extraction module in YOLOv8. MFEM, MFFN, and DNWD, three innovative modules work synergistically to significantly improve the model's detection performance, especially for challenging small objects in remote sensing imagery. The integrated approach addresses key challenges in feature extraction, multiscale fusion, and detection optimization specific to remote sensing imagery analysis.

1) The proposed architecture uses an MFEM-enhanced backbone to progressively refine features extracted from the cross stage partial network with 2 convolutions and fusion (*C2f*) module. This refinement process generates optimized multiscale feature maps (P_2–P_4) with enhanced discriminative capabilities. In particular, while the P_2–P_4 layers retain richer spatial information, the P_5 layer, as the highest level feature representation in the backbone network, exhibits greater semantic abstraction at the cost of reduced spatial detail sensitivity. To address this inherent property of deep convolutional features, we strategically integrate a common spatial attention mechanism prior to the MFEM processing stage of the P_5 layer. This design choice effectively compensates for the progressive loss of fine-grained detail sensitivity in high-level feature representations.
2) An MFFN that enables interactive fusion of the P_2–P_5 features.
3) A DNWD-guided detection head for robust bounding box regression.

Although YOLOv8 demonstrates strong performance as a general-purpose detector [17], it has limitations in detecting small targets, particularly in complex distributions. Our detector builds on the strengths of YOLOv8 while specifically addressing these limitations through the proposed architectural enhancements.

A. Multiscale Feature Enhancement Module

In remote sensing imagery for SOD tasks, targets often exhibit considerable diversity in both size and shape. However, traditional backbone networks typically suffer from limited feature extraction capabilities, resulting in insufficient semantic information and a limited receptive field. This makes it difficult to effectively distinguish small objects at different scales. To address these limitations, we propose an MFEM that enhances small object features through multiscale feature enrichment. The MFEM is embedded into the YOLOv8 backbone after each C2f module. This design mitigates the inherent limitations of C2f, such as its limited multiscale fusion capability, which may underperform compared to more advanced fusion structures for extremely small objects.

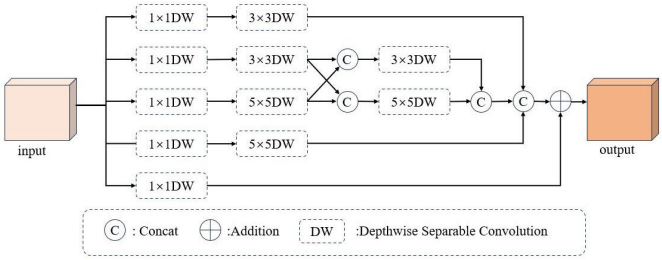


Fig. 2. Structure of MFEM.

The proposed MFEM, shown in Fig. 2, is designed to improve feature representation through a sophisticated multiscale architecture. The module employs a multibranch convolutional structure to enhance feature richness, where different convolutional kernels extract different discriminative semantic information. To expand the receptive field, the MFEM incorporates parallel depth-separable convolutions with different kernel sizes, allowing for comprehensive local contextual information capture. A key innovation of MFEM is its cross-fusion approach, which strengthens feature correlations across scales. The module uses depth-separable convolutions at multiple scales and introduces cross-connections between them, significantly improving the representation of small object features.

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In implementation, the MFEM starts with a 1×1 convolution for channel dimensional adjustment across all branches. The first branch maintains a residual structure with identity mapping to preserve essential small object features. The remaining two branches use 3×3 and 5×5 depth-separable convolutions,[1] respectively, to extract multiscale features while maintaining computational efficiency. After the initial processing, the features undergo cross-fusion, where they are exchanged between branches and further refined by additional 3×3 and 5×5 depth-separable convolutions. This cross-fusion mechanism effectively strengthens feature relationships across scales, resulting in more discriminative representations for challenging detection tasks.

The architecture's emphasis on multiscale processing and cross-branch interaction makes it particularly effective for scenarios requiring fine-grained feature discrimination, such as detecting small objects in complex visual environments. By combining efficient depth-separable convolutions with intelligent feature fusion, the MFEM achieves enhanced performance while maintaining computational efficiency.

B. Multilayer Feature Fusion Neck

In YOLOv8, the neck is a critical component connecting the backbone and the head [17]. Despite its effectiveness in general detection, the neck has some drawbacks in detecting the small targets: 1) the current module performs sequential fusion but lacks strong cross-scale feature interaction that could better enhance the SOD; 2) the fusion process relies on predefined connections that may not optimally adapt to varying object scales in complex scenes; and 3) low-resolution deep features (P_5) may lose fine details, while high-resolution shallow features (P_3) may lack strong semantic meaning, resulting in suboptimal fusion for small targets. To this end, we have developed a novel neck module to replace the original one in Yolov8. The proposed model is called MFFN and is shown in Fig. 3. This clearly demonstrates that the proposed MFFN module redefines the fusion method by systematically combining features across different levels. High-level features capture rich semantic information, while low-level features preserve fine-grained spatial details. Therefore, integrating multiscale feature maps is crucial for improving semantic representation, particularly for SOD.

As shown in Fig. 3, the MFFN takes as an input low-level feature maps (P_2, P_3, P_4) processed by the feature enhancement module and high-level feature map P_5 refined by an attention mechanism. The fusion process uses a hierarchical strategy that progressively integrates features from different scales. The architecture starts by fusing P_4 and P_5 through concatenation and C2f modules to generate the enhanced feature P'_5. This initial fusion establishes a semantic foundation that progressively incorporates finer details through subsequent stages. The architecture then fuses P'_5 with P_3 and P_4 to generate P'_4, effectively bridging the midlevel semantics. Further refinement combines P'_4 with P_2 and P_3 to produce P'_3, capturing the most detailed spatial information. The fusion process continues with deeper integration, where P'_3 and P'_4 interact to produce P''_3. This representation undergoes strategic interaction, refinement, and fusion with adjacent levels to generate P''_4 and P''_5. The final multiscale representations are P''_5, P''_4, and P''_3 for detection tasks.

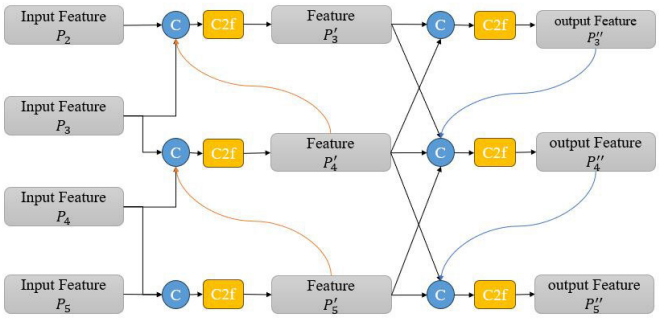


Fig. 3. Structure of MFFN.

In contrast to the traditional neck in the YOLOv8, the advantages of the proposed MFFN module include: incorporation of P_2-layer features to enhance SOD capability, replacement of conventional two-layer fusion with more comprehensive three-layer integration, and hierarchical feature refinement that progressively combines and optimizes multiscale representations. The architecture of the MFFN module introduces novel improvements in multiscale feature integration. The enhanced fusion strategy significantly improves the model's ability to perceive and detect targets at different scales, which is particularly benefiting for SOD in complex remote sensing scenarios.

C. Dynamic NWD

The NWD has emerged as a specialized bounding box loss for SOD that addresses the critical limitations of traditional methods. This innovative approach fundamentally rethinks object representation by modeling bounding boxes as 2-D Gaussian distributions and then quantifying detection accuracy by measuring the similarity between predicted and ground truth distributions. The inherent robustness of the NWD metric to variations in object scale makes it particularly effective for assessing SOD performance. Building on this foundation, our research identifies an important optimization challenge in practical applications. While combining NWD with conventional CIoU has proven effective in frameworks, such as FFCA-YOLO [14] and YOLO-Facev2 [19]. Let L_{NWD} and L_{CIoU} be the loss of NWD and CIoU, respectively. The conventional combined bounding box loss is usually written as follows:

$$\lambda L_{NWD} + (1 - \lambda) L_{CIoU} \tag{1}$$

[1]In terms of selecting the appropriate kernel size, combining 3×3 and 5×5 convolutions strikes an effective engineering balance between modeling local details and contextual awareness. Specifically, odd-sized kernels facilitate better feature alignment: the 3×3 convolution is computationally efficient and economical in terms of parameters, while effectively capturing local structural patterns. The 5×5 convolution, meanwhile, expands the receptive field and enhances contextual feature modeling without significantly increasing computational overhead. In contrast, the even-sized kernels, such as 4×4 or 6×6, offer no clear advantages in terms of feature alignment, hardware optimization, or computational efficiency and are therefore rarely used in mainstream network architectures [18].

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where λ is the parameter to be empirically tuned. The critical issue of weight allocation between these complementary metrics remains unresolved. Current implementations face significant limitations due to static weight configurations, which often lead to suboptimal performance and fail to fully exploit the synergistic potential of both measurement approaches.

To overcome these limitations, we propose DNWD, an adaptive optimization framework that intelligently balances the contributions of NWD and CIoU. This dynamic adjustment mechanism automatically calibrates the weighting based on dataset characteristics and detection requirements, enabling more precise optimization for SOD tasks. The DNWD approach represents a significant advancement over fixed-ratio combinations, offering improved flexibility and performance in diverse detection scenarios. The proposed DNWD introduces an adaptive weighting mechanism that significantly enhances the traditional NWD-CIoU combination. At the core of this innovation is a sophisticated dynamic weighting system that continuously evaluates and adjusts the contributions of both metrics throughout the training process. The weighting mechanism intelligently considers the prediction accuracy at the tth iteration, specifically, the relative accuracy of the NWD-based predictions (p^t) and the CIoU-based predictions (q^t) compared to the ground truth bounding boxes. A relative index Q^t at the tth iteration is defined as follows:

$$Q^t = \frac{p^t}{p^t + q^t}. \tag{2}$$

It is obvious that due to the different characteristics of NWD and CIoU, the Q value will vary with the number of iterations, i.e., there will be prediction variances between different iterations. To automatically seek an optimal balance, the inherent instability during the training phases had to be addressed.

The EMA is a versatile smoothing technique that emphasizes recent data points to effectively filter out noise while preserving underlying trends in time series data. It improves model stability through weight averaging and provides smoothed visualization of training metrics [20]. Inspired by the characteristics of the EMA, we use the EMA to progressively smooth the weighting adjustments, ensuring stable convergence while maintaining the model's responsiveness to meaningful changes in prediction accuracy. In contrast to (1), the tradeoff parameter λ is derived from the EMA. First, a weight given by the EMA technique is written as follows:

$$w_t = 0.9 \times \left(1 - e^{\frac{-t}{c}}\right) \tag{3}$$

where t is the number of iterations, w_t is the weight at the tth iteration, and c is a constant parameter. Generally, the value of c depends on the data size and the total number of iterations [20]. In our model, we empirically set it to 90 000. Using the weight w_t, the tradeoff parameter λ can be derived as follows:

$$\lambda_t = w_t \times Q^t + (1 - w_t) \times \lambda_{t-1} \tag{4}$$

where λ_t denotes the tradeoff parameter at the tth iteration. Obviously, as the number of the iteration increases, the weight w_t becomes closer and closer to 0.9, and the tradeoff parameter λ_t becomes increasingly dependent on the relative index Q^t. Thus, the trade-off parameter λ_t will converge with the increasing number of iterations. Finally, the proposed DNWD is formulated as follows:

$$L_{DNWD} = \lambda_t \times L_{NWD} + (1 - \lambda_t) \times L_{CIoU}. \tag{5}$$

Building on the established relative index Q^t, we incorporate an EMA-based smoothing mechanism. The designed DNWD works by adaptively filtering noise from previous iterations without compromising sensitivity to meaningful performance variations. It continuously adjusts the NWD-CIoU balance in synchronization with the model's evolving detection capability, and it successfully maintains this delicate balance throughout the model's evolutionary trajectory, resulting in consistently robust detection performance. This approach, which is particularly critical for SOD applications, accounts for the inherent variability in feature importance across different detection scenarios and training phases.

III. Experimental Settings

The proposed framework is implemented using PyTorch on an NVIDIA RTX4090 GPU (24-GB VRAM) platform, using CUDA acceleration to maximize computational throughput. We employ stochastic gradient descent optimization with carefully tuned hyperparameters: a momentum setting of 0.937 helps speed up convergence while maintaining training stability, complemented by an initial learning rate of $1e^{-2}$ and weight decay regularization of $5e^{-3}$ to prevent overfitting. The model is trained for 200 epochs with a batch size of 8, striking a balance between memory consumption and gradient estimation accuracy. All experiments use input images at a fixed resolution of 640×640 for training and testing, ensuring consistent evaluation conditions. In subsequent performance comparisons, statistically significant superior results are highlighted in bold for immediate visual identification.

A. Datasets

Two benchmark and challenging datasets are used to evaluate our proposed method. The characteristics of the two datasets are shown in Fig. 4. Additionally, we illustrated the characteristics of the two datasets as follows.

1) *Detection in Optical Remote Sensing Image (DIOR): DIOR* [21] is a large-scale benchmark dataset designed for the detection of small objects in optical remote sensing images. The dataset consists of 23 463 images and 192 472 object instances in 20 categories, including airplanes (APs), airports (APTs), baseball fields (BFs), basketball courts (BCs), bridges (BRs), chimneys (CHs), dams (DAM), highway rest areas (HRAs), highway toll booths (HTBs), harbors (HBs), golf courses (GCs), athletic fields (AFs), overpasses (OPs), ships (SHs), stadiums (STAs), storage tanks (STs), tennis courts (TCs), train stations (TSs), vehicles (VHs), and windmills (WMs). The size of each image is 800×800 pixels, and the spatial resolution varies between 0.5 and 30 m. The DIOR dataset presents numerous challenges for remote sensing SOD. First, targets in these images exhibit significant scale variation (from small VHs to

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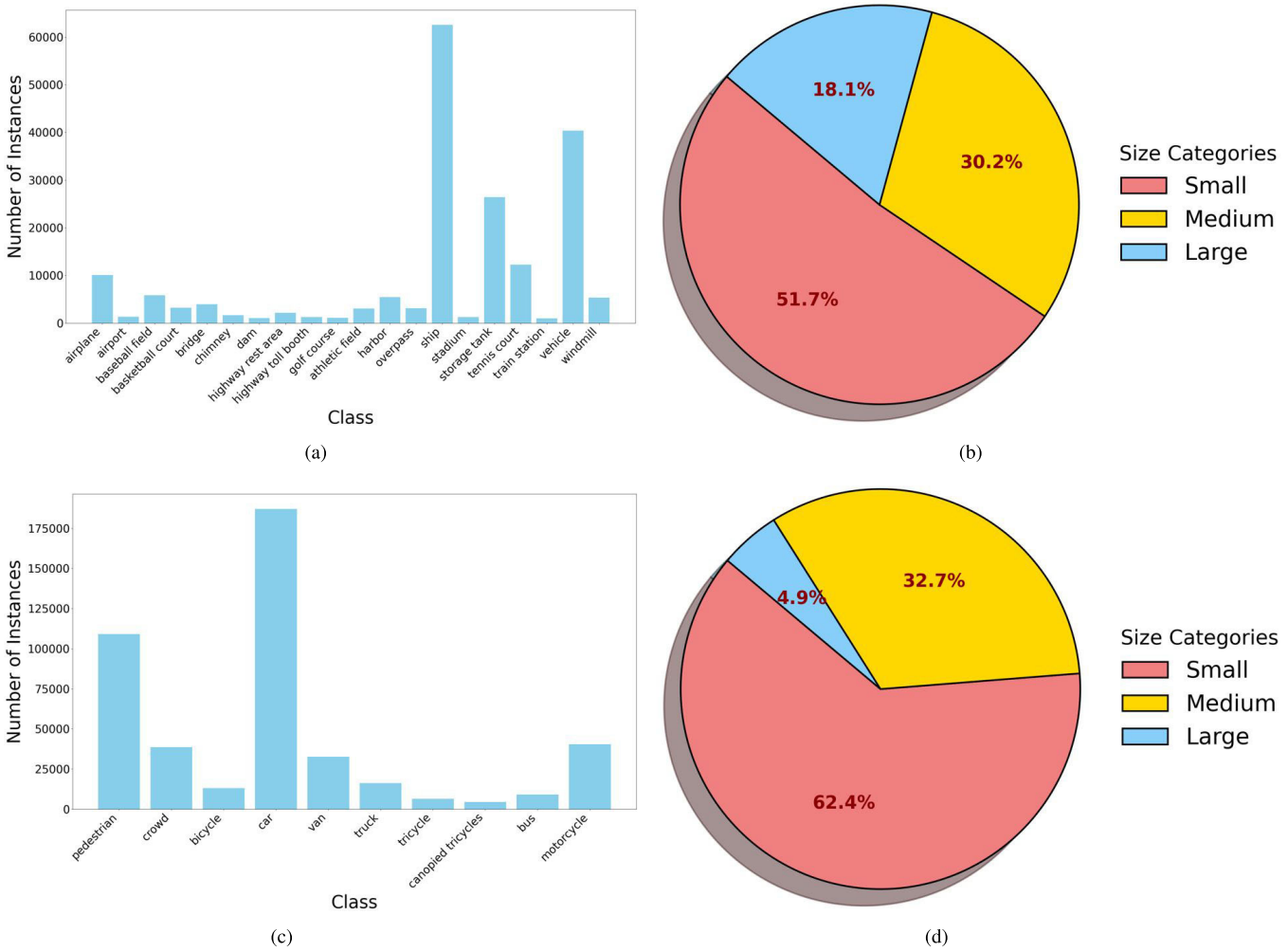


Fig. 4. Object distributions of DIOR and VisDrone2019 datasets. (a) Number of different class objects in DIOR. (b) Objects of different sizes in DIOR. (c) Number of different class objects in VisDrone2019. (d) Objects of different sizes in VisDrone2019.

large buildings), which requires robust multiscale detection capabilities in any effective algorithm. In particular, a large proportion of the objects in the dataset are small (e.g., small VHs, building parts, and wind turbines), characterized by their diminutive size and similar appearance, requiring superior precision in localization and fine-grained discrimination. The distribution of instances across categories and the proportion of objects at different scales, as shown in Fig. 4(a) and (b), underscores this challenge. In addition, the complexity and diversity of the image backgrounds, combined with varying environmental conditions (such as weather and lighting changes), further complicate the detection task. Significant occlusions in some objects and the visual similarity between different categories further exacerbate these difficulties. The datasets were split strictly according to their original official configurations, i.e., the images were divided into 25% for training, 25% for validation, and 50% for testing.

2) *VisDrone2019:* The VisDrone2019 dataset [22], developed primarily by the Visual Intelligence Laboratory at Tianjin University, China, is derived from aerial drone videos taken in several Chinese cities, covering a variety of environments and weather conditions. The officially annotated dataset consists of 6471 images in the training set, 548 images in the validation set, and 1610 images in the test set. This dataset is annotated for ten categories, including pedestrians (PTs), crowds (CWs), bicycles (BIs), cars (CE), vans (VAs), trucks (TUs), tricycles (TYs), canopied tricycles (CY), buses (BUs), and motorcycles (MOs); the distribution of instances across these categories is shown in Fig. 4(c). Notably, as shown in Fig. 4(d), small targets account for a high proportion of the dataset (up to 62.4%), indicating that small targets constitute a significant fraction, even more than half of the dataset. Furthermore, since the dataset is collected from urban and rural environments in 14 cities and includes different weather and lighting conditions, it represents an extremely challenging benchmark for remote sensing SOD.

B. Evaluation Metrics

To thoroughly assess the performance of our proposed method, we use the standard mean average precision (*mAP50*)

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TABLE I
PERFORMANCES OF DIFFERENT VERSION YOLO-BASED MODELS ON DIOR DATASETS

	Params↓	GFLOPs↓	FPS↑	$mAP50\uparrow$	$mAP_S\uparrow$
YOLOv8n	3.0M	8.1	528	66.8	19.1
YOLOv9t [25]	2.6M	10.7	294	67.4	19.9
YOLOv10n [26]	2.7M	8.3	475	66.2	18.5
YOLOv11n [24]	2.6M	6.3	555	67.0	18.2
YOLOv12n [27]	2.6M	6.3	526	66.6	18.4

TABLE II
PERFORMANCE OF THE PROPOSED MODELS WITH DIFFERENT LAYER INPUTS ON THE DIOR DATASET

	Params↓	GFLOPs↓	$mAP50\uparrow$
YOLOv8	3M	8.1	66.8
P2-P4	2.1M	11.5	65.3
P2-P6	5.8M	13.0	67.7
P2-P5	3M	12.2	67.6

metric to evaluate the detection accuracy at a 0.5 IoU threshold. The mAP50 value serves as a direct indicator of detection quality, with improved performance correlating with higher values. Except for mAP50, two marginal metrics precision (P) and recall (R) are used to evaluate the false detection and missed detection, respectively. For computational analysis, we examine both model complexity through the number of parameters (Params) and floating point operations (*GFLOPs*), as well as runtime efficiency through frames per second (*FPS*) measurements. Together, these metrics reveal the fundamental tradeoff between model capability and computational requirements: while increased Params and GFLOPs reflect greater model complexity, higher FPS values indicate superior computational efficiency during inference.

IV. EXPERIMENTS AND ANALYSIS

Using the same default seeds and parameters in the source code ensured that our algorithm and the algorithms we compared it with produced consistent training and testing results on the same dataset. Inspired by the bootstrapping method for confidence interval (*CI*) analysis [23], we modified the original DIOR and VisDrone2019 datasets by applying random horizontal and vertical flips, as well as varying the lighting intensity, to validate the robustness and stability of our algorithm. We then shuffled and recombined the resulting distorted training, testing, and validation sets to create five new datasets, each of which matched the original validation set in terms of class distribution and sample size. We computed the model's mAP50 on these five new datasets and used these values to report the mean mAP50 (denoted by $\overline{mAP50}$), along with its 95% CI, which was calculated based on the t-distribution.

A. Selection of a Baseline Model

To identify the most appropriate baseline model, we performed a comprehensive comparative analysis of different YOLO architectures on the DIOR dataset. The experimental results, presented in Table I, show the performance of the different model versions. As shown in Table I, the detection accuracy, computation complexity, and computation efficiency are considered.

The comprehensive results in Table I reveal an important performance tradeoff between the YOLO variants. Although YOLOv9 has a 0.6% mAP50 advantage over YOLOv8, this improvement comes at a significant computational cost. Compared to YOLOv8, YOLOv11 performs slightly better in both detection accuracy and computational efficiency, as the mAP50 and FPS values in Table I show that YOLOv11n slightly outperforms YOLOv8n. However, the results in the last column[2] of Table I reveal a significant weakness in their ability to accurately detect small objects. Evaluating small-object detection mAP_S across baseline models shows that YOLOv11 significantly underperforms in this critical regard. Furthermore, the YOLOv11's architecture has inherent limitations for detecting small objects. The model's feature extraction and fusion mechanisms may inadequately preserve fine-grained detail, while its default anchoring strategy and postprocessing methods tend to favor larger targets. This architectural bias leads to suboptimal representation and localization of small objects [24].

After a comprehensive evaluation of the experimental results and analysis of the architectural differences between the YOLO versions, we conclude that YOLOv8 provides a superior balance between detection accuracy and computational efficiency. While newer variants, such as YOLOv11, achieve slightly higher mAP50 scores, the inherent limitations of YOLOv11 prove particularly challenging to overcome. Therefore, although YOLOv8 is not the most recent version, we ultimately selected YOLOv8n as our baseline detector.

B. What Is the Best Choice for the Input Layer in MFFN?

We conduct a series of experiments using feature maps from various hierarchical levels to determine the optimal configuration of the input layer for the MFFN. Three input schemes were evaluated: P2–P6, P2–P5, and P2–P4, as summarized in Table II. The experimental results show that the P2–P4 scheme, which excludes the high-level feature map P5, reduces the model's ability to detect blurred objects and causes a significant decrease in overall detection performance. In contrast, the P2–P6 scheme incorporates additional higher-level features that introduce redundancy, yielding only marginal accuracy improvements while substantially increasing the parameter count. Based on these findings, we adopted the P2–P5 scheme for the MFFN because it effectively leverages low-, mid-, and high-level features to achieve accurate, multiscale object detection while maintaining a balance between precision and computational efficiency.

C. Regarding the Setting of the c Value in DNWD

According to (3) and (4), a larger c value requires more training iterations to reach the dynamic equilibrium of (5). Therefore, c primarily governs the rate at which the dynamic weighting parameter converges. The effective convergence of

[2]To specifically evaluate the ability to detect tiny targets, we analyze mAP_S values (mAP50 for targets smaller than 32×32 pixels).

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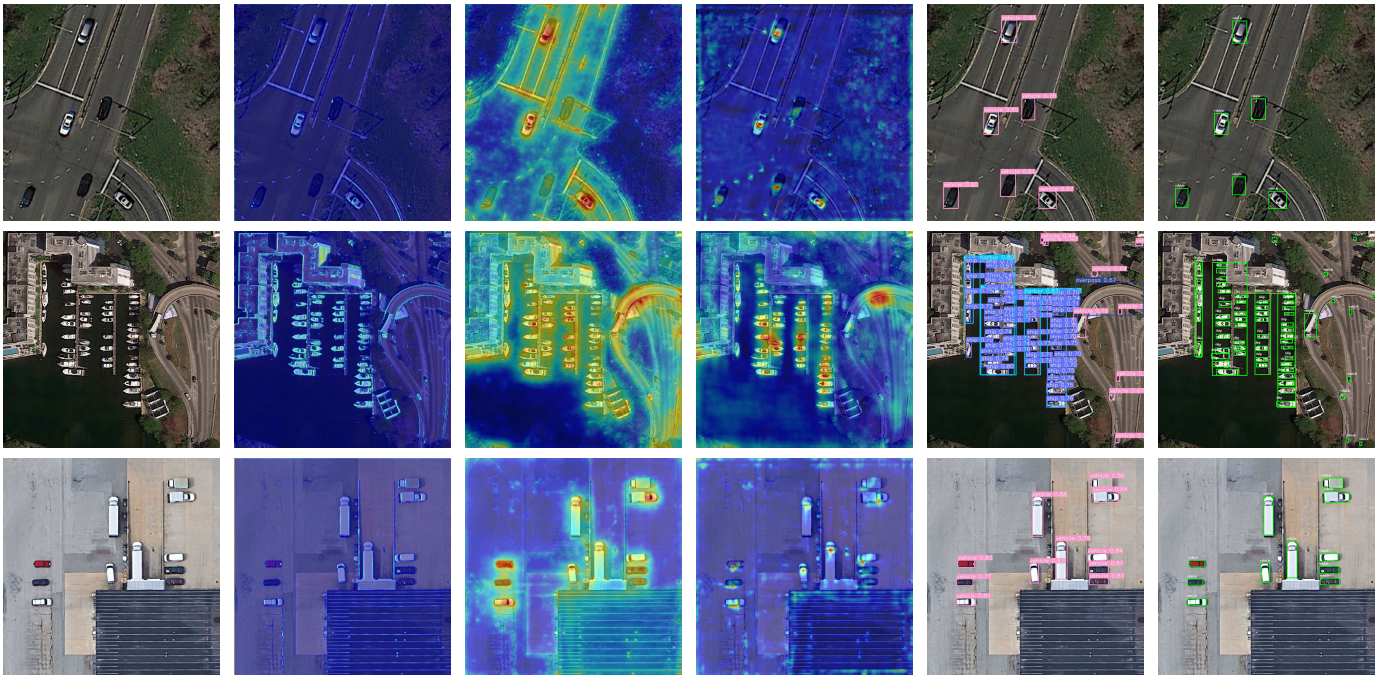


Fig. 5. Grad-CAM visualization results illustrate the impact of MFEM and MFFN on feature extraction at different stages. Warmer colors (closer to red) indicate higher attention from the model to the corresponding regions. The more concentrated the attention on the target, the better the feature extraction performance and the more accurate the model's focus.

TABLE III

PERFORMANCE OF YOLOV8N WITH DIFFERENT c-VALUED DNWD MODULES ON THE DIOR AND VISDRONE2019 DATASETS

	Params↓	GFLOPs↓	DIOR($mAP50$) ↑	VisDrone2019($mAP50$) ↑
YOLOv8	3M	8.1	66.8	26.8
c=80000	3M	8.1	67.2	26.9
c=85000	3M	8.1	67.5	26.9
c=90000	3M	8.1	67.5	26.9
c=95000	3M	8.1	67.2	26.9
c=100000	3M	8.1	67.1	26.9

the dynamic function only occurs once the NWD and CIoU metrics have stabilized. This is influenced by the scale of the dataset and the total number of iterations. In other words, when setting the c value, consideration should be given to these two factors, which is consistent with the theory in [20].

To evaluate the impact of different c values experimentally, we ran a series of experiments within the YOLOv8n framework, each time using a DNWD module with a different c value. The corresponding results are summarized in Table III. According to Table III, the model achieved optimal performance after around 90 000 iterations. Furthermore, the results demonstrate that the DNWD approach outperforms the original YOLOv8n consistently across all tested configurations. Based on the experimental results in Table III for the DIOR and VisDrone2019 datasets, the c value was empirically set to 90 000.

D. Ablation Experiment

To systematically evaluate the impact of each architectural component on SOD performance, we perform a series of ablation studies using the DIOR dataset. Starting from our baseline model, we sequentially integrated the MFEM, the MFFN, and the DNWD module and evaluated their individual and collective contributions through extensive testing on the DIOR dataset. The quantitative results are shown in Tables IV and V, where "√" represents using the module and "×" represents not using the module. We also show the extracted feature maps and detection maps with the comparisons in Fig. 5. Obviously, the ablation studies systematically demonstrate the complementary benefits of each proposed SOD module.

The results in the second row of Table IV show that the MFEM module outperforms the baseline model YOLOv8 a 0.3% mAP50 improvement with negligible computational overhead and the feature visualizations in Fig. 5 confirming

TABLE IV

ABLATION EXPERIMENTAL RESULTS ON DIOR DATASET WITH BASELINE MODEL YOLOV8N

MFEM	MFFN	DNWD	Params↓	GFLOPs↓	$\overline{mAP50}$ ↑	95%CI
×	×	×	3.0M	8.1	71.2	[69.0 73.4]
√	×	×	3.0M	8.1	71.5	[69.3 73.7]
×	√	×	2.9M	8.9	71.8	[69.8 73.8]
×	×	√	3.0M	8.1	71.5	[69.5 73.5]
√	√	×	3.2M	8.9	72.0	[70.0 74.0]
√	×	√	3.0M	8.1	72.1	[69.9 74.3]
×	√	√	3.2M	8.9	71.9	[70.0 73.8]
√	√	√	3.2M	8.9	**72.8**	**[70.9 74.7]**

TABLE V

ABLATION EXPERIMENTAL RESULTS ON DIOR DATASET WITH BASELINE MODEL YOLOV11

	MFEM	MFFN	DNWD	Params↓	GFLOPs↓	$\overline{mAP50}$ ↑	95%CI
YOLOv11	×	×	×	2.6M	6.3	71.6	[69.3 73.9]
	√	×	×	2.8M	7.2	71.9	[70.0 73.8]
	√	√	×	3.9M	11.5	72.2	[70.3 74.1]
	√	√	√	3.9M	11.5	**72.6**	[70.7 74.5]

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its ability to improve target-background discrimination through enriched local context. Compared to the MFEM module, the MFFN module individually achieves more substantial gains while actually reducing the number of parameters, validating its effectiveness in exploiting critical low-level features through optimized multiscale fusion. While the DNWD module makes a unique contribution to localization accuracy, compared to the baseline model YOLOv8, increasing mAP50 by 0.6% through its adaptive weighting of geometry-sensitive metrics. Importantly, these modules exhibit strong synergistic effects: the combination of MFEM and MFFN yields 72.0% mAP50, while the full integration of all three components establishes a new advanced performance of 72.8%, an absolute improvement of 1.6% over the baseline. As summarized in Table IV, our complete method achieves a mAP50 of 72.8% with a 95% CI of [70.9%, 74.7%]. This interval is considerably narrower than those of the baseline and all the ablation variants, which underscores the stability of our approach. Furthermore, the lower and upper bounds of our method's CI are higher than those of all competing algorithms, offering robust statistical evidence that our algorithm significantly outperforms others in SOD tasks.

In particular, as shown in Fig. 5, the MFEM can effectively localize regions of interest when compared to the original feature maps obtained by a common convolutional layer. Based on these localized regions, the MFFN successfully extracts small target features from complex backgrounds. Finally, in contrast to the ground truth, the DNWD accurately detects densely distributed small targets through its adaptive bounding box adjustment mechanism.

To further evaluate the proposed modules, we perform ablation experiments using YOLOv11 as the baseline model on the DIOR dataset. The experimental results presented in Table V corroborate the results in Table IV, demonstrating a consistent improvement in accuracy with our module implementation. Significantly, the full integration of all three designed modules enables YOLOv8 to outperform YOLOv11 in overall accuracy with equally wide CIs, highlighting its superior compatibility and scalability. Overall, these results show that the MFEM, MFFN, and DNWD modules work synergistically to achieve significant performance gains in SOD. These modules complement each other's functions and together improve detection robustness without compromising computational efficiency. This represents a significant advancement over existing solutions for SOD in complex remote sensing scenarios.

In other words, in contrast to the baseline models, the results in Tables IV and V and Fig. 5 demonstrate that our proposed approach effectively overcomes the inherent challenges of the DIOR dataset, including multiscale small objects, complex backgrounds, target occlusion, and high interclass similarity. This is achieved through enhanced cross-scale information interaction, refined fine-grained feature extraction, and optimized adaptive bounding box design.

E. Comparison Experiments

To perform the comparison experiments, we selected several recently published small object detectors. The following is a brief overview of these methods along with their main characteristics.

1) *Faster-RCNN [28]:* A classic two-stage detector that incorporates a region proposal network to quickly generate candidate regions. Its unified, end-to-end training pipeline provides significant gains in both inference speed and detection accuracy.
2) *YOLOv9 [25]:* Developed by Wang et al. [25] on top of YOLOv7, this variant introduces programmable gradient information to provide richer task-specific inputs, resulting in more reliable gradient signals for network weight updates.
3) *YOLOv11 [29]:* Released by Ultralytics, this iteration refines the C2f module from YOLOv8, enhances the cross stage partial backbone, and embeds a pyramid slice attention mechanism. Its revamped spatial pyramid pooling fast block merges global average pooling with global max pooling to better capture global context.
4) *FFCA-YOLO [14]:* Builds on the standard YOLO architecture by integrating a feature enhancement module, a feature fusion module, and a spatial context awareness module. Through multiscale feature fusion, it enhances the semantic representation of small objects and suppresses background clutter, significantly improving localization and classification in remote sensing imagery.
5) *TPH-YOLO [30]:* Designed for drone-captured scenes with extreme scale variation, dense occlusions, and complex backgrounds. It extends the multiscale detection heads, uses a transformer for comprehensive feature modeling, and optimizes the attention mechanism to increase robustness.
6) *Gold-YOLO [31]:* From Huawei Noah.s Ark Lab, this model's hallmark is the gather-and-distribute mechanism, which enhances multiscale feature fusion, striking a superior balance between detection latency and accuracy.

To maintain experimental rigor and ensure fair comparisons, all baseline models were carefully implemented using their original open-source codes without the use of pretrained weights. In particular, we standardized model architectures by scaling TPH-YOLO to match YOLOv5s in both depth and width dimensions. For a comprehensive evaluation, we further validated our approach across multiple model scales, including the "n" and "s" variants, with special attention to size equivalence (e.g., the "t" size of YOLOv9 corresponds to the "n" size of other models). Specifically, the terms "Ours-n" and "Ours-s" refer to the detectors we have developed, which are based on YOLOv8n and YOLOv8s, respectively. The detailed comparison results on the DIOR and VIsDrone2019 are systematically listed in Tables VI and VII, respectively.

The experimental results in Table VI demonstrate the superior precision, recall, and mAP values of our method compared not only to baseline models of equivalent scale across all configurations but also to the several recent small object detectors. The DIOR dataset contains 20 target categories, and our proposed algorithm (Ours-n) demonstrates superior detection rates for 12 of these classes, including the most

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TABLE VI

PERFORMANCE OF DIFFERENT ALGORITHMS ON DIOR DATASET

	$Params$	$GFLOPs$	FPS	AP	APT	BF	BC	BR	CH	DAM	HRA	HTB	HB	GC	AF	OP	SH	STA	ST	TC	TS	VH	WM	P	R	$\overline{mAP50}$	95%CI
YOLOv8n	3M	8.1	528	81.1	75.2	77.7	86.7	35.9	75.5	57.0	57.2	47.9	63.2	72.3	65.9	54.9	87.7	65.8	68.6	87.6	44.0	44.6	**76.2**	82.4	59.6	71.2	[69 73.4]
YOLOv9t	2.6M	10.7	294	**86.5**	70.2	77.9	**87.5**	36.3	76.0	48.0	57.8	48.3	61.3	71.6	68.4	55.7	87.1	**73.4**	**70.8**	**89.3**	49.6	42.8	76.1	81.0	60.9	71.8	[69.6 74]
YOLOv11n	2.6M	6.3	555	80.8	75.6	76.1	86.7	33.2	75.7	57.7	56.9	47.5	62.8	75.2	65.4	53.7	88.3	66.5	70.5	87.7	48.1	45.2	74.1	82.6	59.6	71.6	[69.3 73.9]
Gold-YOLOn	5.61M	12.07	436	76.3	72.7	72.6	85.4	34.6	73.5	51.7	**59.3**	**53.0**	63.0	73.7	62.5	54.5	87.7	60.4	66.0	85.4	53.8	43.1	73.8	82.6	58.8	70.6	[68.7 72.5]
Ous-n	3.2M	8.9	476	81.4	**77.4**	**78.4**	87.2	**37.1**	**76.1**	**58.1**	57.9	47.8	**63.8**	**75.9**	**70.2**	**56.9**	**88.4**	72.6	70.7	88.0	**56.7**	**46.2**	76.0	**82.8**	**61.7**	**72.8**	[70.9 74.7]
YOLOv8s	11.1M	28.5	456	81.3	78.7	78.2	88.4	41.2	77.4	58.8	57.6	50.2	63.6	77.0	72.2	57.7	90.1	72.2	75.6	89.0	52.5	50.8	78.4	83.4	62.8	74.7	[72.6 76.8]
YOLOv9s	9.6M	38.8	246	**84.0**	80.1	79.0	89.4	42.0	**79.0**	**63.5**	57.2	48.5	65.1	78.0	73.5	58.9	89.7	71.1	75.2	89.7	59.7	51.6	77.8	83.3	63.6	76.2	[73.8 78.6]
YOLOv11s	9.4M	21.3	542	83.1	**81.7**	77.7	89.4	40.1	77.0	58.9	57.1	50.1	63.3	76.7	72.3	57.5	89.9	72.0	73.7	88.8	52.2	49.9	78.1	83.5	62.7	74.9	[72.6 77.2]
Faster-RCNN	41.5M	134	55	77.0	56.4	76.7	88.4	35.8	78.4	40.8	57.1	51.0	46.7	68.6	73.3	52.7	82.8	57.1	73.4	86.9	40.7	49.3	77.1	76.1	54.3	67.9	[63 72.8]
Gold-YOLOs	21.5M	46	377	77.3	72.0	73.2	86.3	34.5	74.5	58.5	59.2	52.6	62.6	74.1	63.5	54.3	88.4	60.7	65.3	86.3	52.4	44.9	75.3	83.6	58.9	73.5	[69.3 77.7]
FFCA-YOLO	7.2M	51.7	181	82.0	52.8	**80.1**	87.9	35.2	75.8	29.7	50.6	57.4	53.7	57.7	65.9	48.9	90.5	73.1	**78.4**	88.2	31.7	55.5	74.9	71.6	60.6	74.0	[68.9 79.1]
TPH-YOLO	7.2M	30	370	81.4	66.0	77.9	86.9	37.1	75.7	47.4	**61.2**	**58.0**	56.9	69.4	66.6	54.2	88.4	61.5	74.4	86.6	45.8	51.2	74.7	80.3	60.0	70.8	[68.5 73.1]
Ous-s	13.9M	34.8	417	82.5	80.5	77.9	**89.6**	**43.0**	78.3	60.5	57.8	49.8	**65.5**	**79.5**	**74.1**	**58.9**	**90.8**	69.4	76.5	**90.1**	**61.6**	**55.7**	**79.9**	**83.6**	**64.0**	**76.5**	[74.1 78.9]

TABLE VII

PERFORMANCE OF DIFFERENT ALGORITHMS ON VISDRONE2019 DATASET

	$Params$	$GFLOPs$	FPS	PT	CW	BI	CE	VA	TU	TY	CY	BU	MO	P	R	mAP_S	$\overline{mAP50}$	95%CI
YOLOv8n	3M	8.1	517	22.3	12.1	5.77	66.0	30.4	29.5	12.1	14.2	50.1	23.6	39.0	28.9	13.5	34.8	[30.8 38.8]
YOLOv9t	2.6M	10.7	283	20.8	11.0	6.2	66.4	**31.6**	33.3	11.0	13.5	**52.2**	21.6	38.8	29.3	13.4	33.6	[30.3 36.9]
YOLOv11n	2.6M	6.3	532	21.4	11.5	5.7	65.4	29.7	30.0	12.0	13.8	49.7	23.2	36.2	29.8	13.1	34.1	[30.3 37.9]
Gold-YOLOn	5.61M	12.07	421	20.5	10.3	5.7	66.2	31.5	31.6	10.9	**14.9**	50.1	21.7	37.5	29.3	13.4	31.8	[29.0 34.6]
Ous-n	3.2M	8.9	449	**22.8**	**13.2**	**6.2**	**66.6**	31.5	**33.5**	**12.8**	14.6	51.7	**28.3**	**39.5**	**30.5**	**14.1**	**36.9**	[32.4 40.4]
YOLOv8s	11.1M	28.5	451	26.0	13.5	7.87	70.4	36.4	38.4	17.3	16.3	54.2	27.9	42.7	32.9	16.7	42.8	[36.9 48.7]
YOLOv9s	9.6M	38.8	238	26.3	13.7	9.73	71.6	38.2	40.3	17.6	**21.8**	**58.3**	29.0	46.0	34.5	17.8	45.0	[39.0 51.0]
YOLOv11s	9.4M	21.3	532	25.1	13.8	8.62	70.3	37.4	38.6	17.7	18.3	56.7	26.9	45.7	32.6	17.2	43.6	[38.2 49.0]
Faster-RCNN	41.5M	134	52	25.5	13.1	9.1	70.8	37.6	38.4	**21.4**	19.0	56.0	26.1	45.9	33.6	18.8	42.8	[37.4 48.2]
Gold-YOLOs	21.5M	46	362	22.4	10.7	6.8	68.9	34.5	33.4	12.0	16.9	53.4	24.3	40.2	31.2	15.0	35.9	[31.7 40.1]
FFCA-YOLO	7.2M	51.7	175	**27.5**	13.5	9.8	**75.3**	38.3	38.9	17.6	20.5	57.3	29.1	43.9	34.6	18.2	45.5	[39.3 51.7]
TPH-YOLO	7.2M	30	355	24.5	13.7	8.24	67.7	26.9	23.4	10.4	9.46	49.0	22.8	34.1	30.6	17.3	34.2	[30.0 38.4]
Ous-s	13.9M	34.8	412	26.4	**14.6**	**10.5**	72.6	**39.1**	**40.8**	18.6	20.6	58.0	**29.2**	**46.2**	**34.7**	**21.0**	**45.7**	[39.5 51.9]

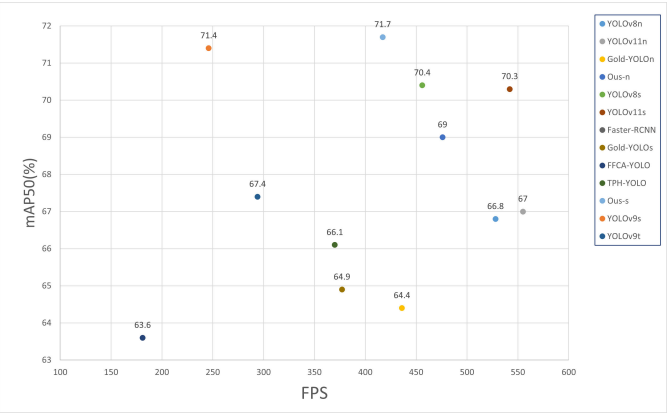


Fig. 6. FPS and mAP50 distribution of different models on the DIOR dataset. The horizontal axis represents FPS, where higher values indicate faster processing speed. The vertical axis represents mAP50, where higher values indicate better detection accuracy.

common small targets (ships), the second most common small targets (vehicles), and the windmill (WM) with an average proportion, outperforming all competing methods. Specifically, our approach achieves mAP50 values of 72.8% with a 95% CI of [70.9% 74.7%] and 76.5% with a 95% CI of [74.1% 78.9%] for the compact and standard versions, respectively, representing significant improvements over competing methods. Although the upper bound of our algorithm's 95% CI is slightly lower than that of FFCA-YOLO, our CI is substantially narrower. Consequently, our method's entire performance range, from worst case to best-case estimates, surpasses all other algorithms' best-case estimates. This demonstrates that our algorithm's performance is statistically and unequivocally superior to all comparative methods, even when accounting for random variation. The performance advantage is particularly pronounced for small target categories, such as vehicles, windmills, and ships, while maintaining strong detection capability for medium to large objects, such as basketball courts and bridges.

Fig. 6 illustrates the correlation between FPS and mAP50 across all the methods that were compared on the DIOR dataset. Although our model has a slightly higher parameter count and computational load than YOLOv8s due to its advanced feature aggregation and extraction mechanisms, it substantially improves detection accuracy. This tradeoff aligns with common practices in the SOD task. Furthermore, compared to lower FPS models, such as FFCA-YOLO, our approach maintains a clear accuracy advantage while operating at higher inference speeds. This highlights its favorable position in the accuracy–efficiency balance.

On the more challenging VisDrone2019 dataset, our proposed methods demonstrate superior performance. As shown in Table V, Ours-n and Ours-s achieve mAP50 values of 36.9% and 45.7%, respectively, representing significant improvements of 5.2% and 11.6% over Gold-YOLOn and TPH-YOLO. Compared to the DIOR dataset, the VisDrone2019 dataset contains more challenging small objects. This results in a decrease in the mean average precision at mAP50 and an increase in the width of the 95% CI. However, our algorithm achieves a significantly higher average mAP50 than other algorithms and demonstrates higher upper and lower bounds of the 95% CI. Our detector achieves a remarkable mAP_S of 21% for sub-32 $\times$ 32 pixel targets, outperforming Gold-

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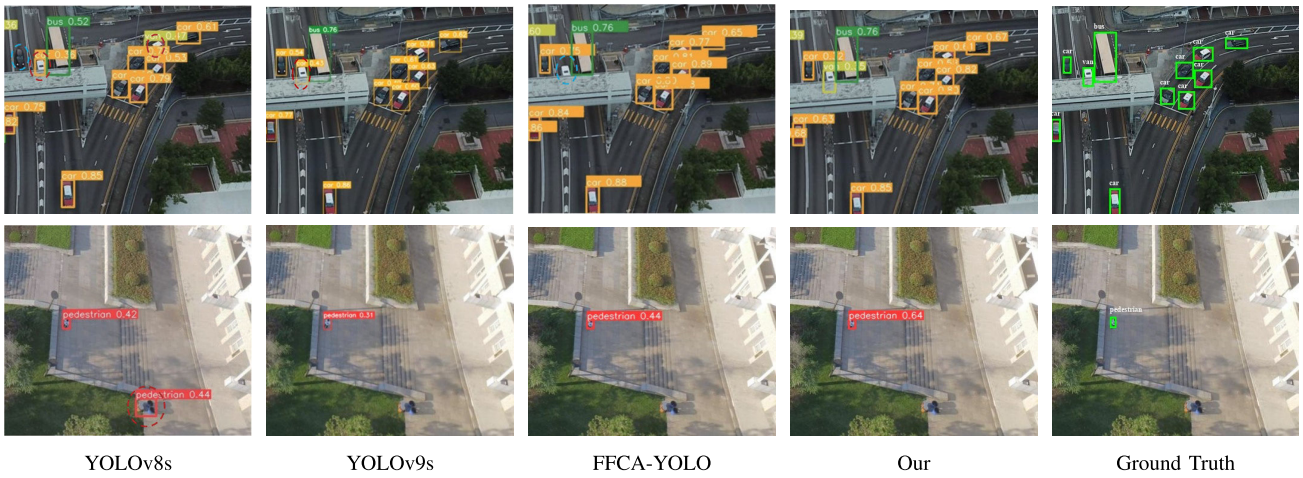


Fig. 7. Detection maps using YOLOv8s, YOLOv9s, FFCA-YOLO, and our method on images in VIsDrone2019 dataset.

YOLOs by 6%. Notably, it delivers the highest detection rates in most categories, with particularly outstanding performance in challenging cases, such as crowds (CW) and bicycles (BI). These results robustly validate the effectiveness and superiority of our approach for SOD tasks.

To visually demonstrate the detection performance, Fig. 7 compares the detection maps on the VisDrone2019 dataset between our proposed method and three representative methods. For clear error visualization, false alarms and misdetections are marked with red ellipses, while missed detections are marked with blue ellipses. The proposed method shows excellent agreement with the ground truth data, with significantly fewer detection errors compared to the other three methods compared, while the other compared methods show different combinations of false alarms, missed detections, and misdetections, e.g., YOLOv8s produced one false alarm in pedestrian detection and one false and one missed detection in vehicle detection. YOLOv9s and FFCA-YOLO each produced one false alarm and one missed detection. This visual evidence strongly supports the improved performance of our detector design and validates the effectiveness of the proposed architectural modules.

The above experimental results and analysis show that our proposed detector consistently achieves the highest detection rates despite the numerous challenges present in both the DIOR and VisDrone2019 datasets, including complex and variable scene backgrounds, diverse target sizes, and dense distributions of potentially occluded small targets. This superior performance is due to our innovative architectural design. The proposed MFEM effectively extracts and characterizes regions of interest from cluttered backgrounds through multiscale feature interaction and fusion. Building on this, the MFFN module captures subtle yet discriminative features through multilevel information interaction, while the adaptively adjusted bounding boxes significantly improve the anchor quality. The synergistic operation of these three components together enhances the detection accuracy for small targets.

It should be noted that, as shown in Fig. 6 and Tables VI and VII, our method requires more parameters and computational resources compared to the YOLO series. However, it maintains lower computational complexity and higher efficiency than several other state-of-the-art SOD methods. In particular, our algorithm achieves FPS values that are 236 and 237 higher than FFCA-YOLO on the DIOR and VisDrone2019 datasets, respectively, demonstrating its practical advantages in real-time applications.

V. Conclusion

This article presents a robust detector specifically re-engineered for remote sensing SOD with dense distribution and clutter environments. To improve the feature extraction capability for small clutter objects, we have developed the MFEM and the MFFN module. The MFEM uses multiple depth-separable convolutions at different scales to improve the network's perception of objects of different sizes, while enhancing feature correlations between scales by cross-fusing convolution outputs, resulting in richer local contextual information capture. The MFFN enhances the neck's feature fusion capability by incorporating additional lower layers containing small object features and integrating multilevel features, thereby enriching the hierarchical contextual representation of the feature pyramid. Complementing this, the MFFN redefines the neck architecture, which significantly enriches the hierarchical representation of the feature pyramid. Furthermore, we introduce the DNWD loss, which utilizes an EMA-based dynamic weighting mechanism to adaptively balance NWD and CIoU. This innovation dynamically adjusts the NWD weight, allowing the loss function to more accurately capture subtle differences in object shape and scale during bounding box regression for small objects. Extensive experiments on the DIOR and VisDrone2019 datasets demonstrate that our approach not only achieves superior overall accuracy but also delivers exceptional performance on tiny targets (mAP_S).

The proposed method demonstrates significant performance improvements on the DIOR and VisDrone2019 datasets. However, there are several limitations that must be addressed in future work. While the proposed algorithm outperforms other methods on representative and challenging SOD datasets and has higher mAP values than other algorithms for most categories, including large, medium, and small objects, its performance is inferior for certain categories. This is due to

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imbalanced training sample distribution and the algorithm's stronger focus on handling complex, small objects. Second, our approach introduces additional parameters and computational overhead compared to baseline models, such as YOLOv8n/s. Future research will focus on exploring efficient self-supervised learning and lightweight architectural designs to further improve the model's generalization and deployment efficiency across diverse remote sensing application scenarios.

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